

Emily Tran

Los Angeles, CA • (805) 248-9211 • emilyclaratran@yahoo.com • ectran.github.io/PortfolioWebsite

Education

California State Polytechnic University, Pomona

Bachelor of Science in Computer Science, Minor in Data Science

GPA: 3.74 | Magna Cum Laude

Aug 2022 – May 2026

Pomona, CA

Projects

Gambler's Addict | *Unity, C#, Gameplay Systems, Adaptive AI*

- Designed and implemented adaptive gameplay, economy, and progression systems for a 2D bullet-hell roguelike
- Engineered modular AI behaviors and dynamic difficulty scaling systems
- Developed debugging and gameplay visualization tools to support balancing and playtesting iteration
- Built persistent progression systems including save-state management, cross-scene data persistence, and adaptive reward multipliers

Funk vs Fright | *Unity, C#, Animation Systems, Gameplay Systems*

- Led a 4-person interdisciplinary team in coordinating gameplay, art, and UI development for a 2.5D rhythm-combat game
- Implemented timing-sensitive gameplay systems, animation state management, and gameplay feedback loops
- Created UI assets, sprite animations, and visual feedback systems to improve gameplay readability
- Awarded 1st Place overall in a competitive game jam

Calendar Meet | *React, Node.js, PostgreSQL, AWS, Github Actions*

- Collaborated on the development of a full-stack scheduling platform using React, Node.js, and PostgreSQL
- Configured AWS EC2 deployment infrastructure and implemented CI/CD pipelines using GitHub Actions
- Maintained application deployments, database integrations, and production updates

Experience

Virtual Reality Lab — Cal Poly Pomona

Mar 2025 – Jun 2025

Unity Developer

Pomona, CA

- Developed VR object interactions and gameplay mechanics using the XR Interaction Toolkit.
- Implemented enemy chase and attack AI behaviors and integrated spatial footstep audio to improve immersion
- Tested, debugged, and refined gameplay systems through iterative development and user feedback
- Collaborated with a four-person Agile development team using Git for version control

Research Experience for Undergraduates (REU) — Texas A&M University

May 2024 – Aug 2024

Applied Computational Robotics Researcher

College Station, TX

- Developed Python simulations to evaluate autonomous robot navigation in dynamic environments
- Implemented and compared A* and D* Lite pathfinding algorithms under changing obstacle scenarios
- Benchmarked algorithm performance by analyzing path quality, replanning behavior, and computational efficiency
- Presented research findings and documented experimental methodology and results

Technical Skills

Languages: C#, Python, Java, JavaScript

Frameworks & Technologies: Unity, React, Node.js, PostgreSQL, AWS

Game Development: Gameplay Systems, Game AI, XR Interaction Toolkit

Developer Tools: Git, GitHub, GitHub Actions, VS Code, Aseprite